Calculations & Formulas Reference

Pre-Flight Shipment Checker · every number the app computes, the exact formula, and what changes when an input changes mid-way

For Claude Code: every formula here is implemented in `calculations.ts` (function names in the last column of each table) and covered by `calculations.test.ts`. Do not re-implement these formulas elsewhere, and never let Gemini compute them. Companion files: RULEBOOK.md, RECOMMENDATION_RULES.md, QUESTIONS.md.

1. Conventions

Topic	Convention
Money	All GBP. Products and sums computed in whole pence (×100, rounded) to avoid floating-point error, shown with 2 decimals (£5,400.00).
Weights	kg. Item weights to 3 dp; weight per pallet to 2 dp. Grams ÷ 1000, tonnes × 1000 when parsing.
Percentages	Shown to up to 3 dp. Tolerance comparisons done in pence, not percentages.
Comparisons	Limits are inclusive: “must not exceed” = ≤, “at least” = ≥, “within 1%” = ≤ 1%. Equal-to-limit always passes.
Whole numbers	Pallets, cartons, quantity and units per carton must be whole numbers ≥ 1 (else input error V-05/V-08).
Dates	Stored as YYYY-MM-DD, displayed DD-MM-YYYY with weekday. “Today” = current date in Europe/London.
Working days	Monday–Friday; no bank holidays. Config TRANSIT_DAY_MODE = business (alternative: calendar).
Notation	Σ = sum over all items of the booking; ⌈x⌉ = round up to whole number; |x| = absolute value; mod = remainder.

2. Item-level formulas (per row of Sheet 2)

ID	Quantity	Formula	Example (MUG-01)	Function
F-I1	Line value (£)	quantity × unit price	1,080 × £5.00 = £5,400.00	`itemLineValue`
F-I2	Net weight (kg)	quantity × unit weight	1,080 × 0.5 = 540 kg	`itemNetWeight`
F-I3	Packed units	units per carton × cartons	12 × 90 = 1,080	`itemPackedUnits`
F-I4	Units difference	packed units − quantity	1,080 − 1,080 = 0	`itemUnitsDiff`
F-I5	Cartons needed	⌈quantity ÷ units per carton⌉	⌈1,080 ÷ 12⌉ = 90	`itemCartonsNeeded`
F-I6	Exact cartons	quantity ÷ units per carton, only if quantity mod units per carton = 0	90	`itemExactCartons`
F-I7	Units in last carton	quantity mod units per carton (0 = all cartons full)	0	`itemUnitsInLastCarton`
F-I8	Net weight per carton	units per carton × unit weight	12 × 0.5 = 6 kg	`itemCartonNetWeight`
F-I9	Value per carton	units per carton × unit price	12 × £5 = £60.00	`itemCartonValue`
F-I10	HS digits	HS code with spaces, dots, dashes removed; count digits	691200 → 6	`normaliseHs`

3. Booking-level formulas (Sheet 1 row + its items)

ID	Quantity	Formula	Example (BK-001)	Function
F-B1	Total units	Σ quantity	1,080	`computeTotals.sumUnits`
F-B2	Computed goods value (£)	Σ F-I1 (in pence)	£5,400.00	`sumItemValue`
F-B3	Σ item cartons	Σ item cartons	90	`sumItemCartons`
F-B4	Net weight (kg)	Σ F-I2	540 kg	`sumNetWeightKg`
F-B5	Declared value difference (£)	declared value − F-B2	5,400 − 5,400 = 0	`declaredValueDiff`
F-B6	Declared value difference (%)	|F-B5| ÷ F-B2 × 100	0%	`declaredValueDiffPct`
F-B7	Value tolerance (£)	1% × F-B2	£54.00	`checkR02`
F-B8	Cartons difference	booking cartons − F-B3	90 − 90 = 0	`cartonsDiff`
F-B9	Weight per pallet (kg)	gross weight ÷ pallets	620 ÷ 3 = 206.67	`weightPerPalletKg`
F-B10	Packaging weight (kg)	gross weight − F-B4	620 − 540 = 80	`packagingWeightKg`
F-B11	Cartons per pallet (avg)	F-B3 ÷ pallets	30	`avgCartonsPerPallet`
F-B12	Average unit value (£)	F-B2 ÷ F-B1	£5.00	`avgUnitValue`
F-B13	Suggested minimum gross (kg)	F-B4 + 15 × pallets (advice only, never auto-filled)	540 + 45 = 585	`deriveBookingTotals`

4. Rule formulas (official rulebook R-01 … R-10)

Rule	Sev.	Pass condition (exact)	Fix calculation shown to the seller	Function
R-01	Red	for every item: HS digits only AND count ≥ 6	Blank/short → ask for full code. 5-digit numeric cell → suggest “0” + code (lost leading zero)	`checkR01`
R-02	Red	|declared − F-B2| ≤ 1% × F-B2 (in pence)	Corrected declared value = F-B2; difference £ and %	`checkR02`
R-03	Red	for every item: units per carton × cartons = quantity	cartons = quantity ÷ upc (if whole); else ⌈quantity ÷ upc⌉ with remainder; quantity = upc × cartons; upc = quantity ÷ cartons (if whole)	`checkR03`
R-04	Red	booking cartons = Σ item cartons	Corrected booking cartons = F-B3; difference = F-B8	`checkR04`
R-05	Red	gross ÷ pallets ≤ max kg/pallet of the chosen carrier	Min pallets = ⌈gross ÷ max⌉ and its cost; cost/ETA of each carrier that fits	`checkR05`
R-06	Red	no item matches a definite lithium / aerosol / flammable keyword or HS prefix (after removing negations), and no “possible” match confirmed by Gemini/seller	Remove item or correct description	`checkR06, detectDangerousGoods`
R-07	Amber	gross ≥ Σ(unit weight × quantity)	Shortfall = net − gross; suggested ≥ net + 15 × pallets	`checkR07`
R-08	Amber	NOT (destination type = Warehouse AND appointment = No)	Book slot on/after ETA	`checkR08`
R-09	Amber	for every item: country of origin not blank	Ask where it was made	`checkR09`
R-10	Amber	ready date ≥ today (UK)	Days in the past = today − ready date	`checkR10`

4.1 Which carrier R-05 uses

- If the seller explicitly chose a carrier → that carrier.
- Otherwise (R05_MODE = priority_pick, default) → the carrier the seller's priority would choose if weight were ignored (among carriers meeting any deadline/budget). This surfaces an overweight problem, e.g. Cheapest → Carrier A (500 kg).
- Alternative (R05_MODE = recommended) → the best eligible carrier; R-05 then fails only if no carrier fits.

4.2 Status

```
status = RED        if any Red issue
       = INCOMPLETE if no Red but any Data issue (missing / unreadable input)
       = AMBER      if only Amber (and Advisory) issues
       = GREEN      otherwise
canConfirm = status in {AMBER, GREEN};  AMBER needs the seller to acknowledge warnings
```

5. Carrier formulas (Sheet 3, per carrier on the route)

ID	Quantity	Formula	Function
F-C1	Freight cost (£)	price per pallet × pallets	`shipmentCost`
F-C2	Fits weight	F-B9 ≤ max kg per pallet	`evaluateCarrier`
F-C3	Minimum pallets	[gross ÷ max kg per pallet]	`minPalletsFor`
F-C4	Max gross for N pallets	max kg per pallet × N	`maxGrossFor`
F-C5	Cost if re-palletised	price per pallet × F-C3 (only when F-C3 > pallets)	`evaluateCarrier`
F-C6	Collection date	start = max(ready date, today); if Sat/Sun → next Monday	`collectionDate`
F-C7	Estimated delivery (ETA)	F-C6 + transit days (Mon–Fri only)	`etaDate`, `addBusinessDays`
F-C8	Deadline	earliest of (required delivery date, appointment date)	`earliestDeadline`
F-C9	Meets deadline	ETA ≤ deadline	`evaluateCarrier`
F-C10	Spare days	deadline − ETA (calendar days)	`recommend`
F-C11	Within budget	F-C1 ≤ max budget	`evaluateCarrier`
F-C12	Late risk (%)	100 − on-time %	`latePct`
F-C13	Cost per kg / per unit	F-C1 ÷ gross; F-C1 ÷ F-B1	`evaluateCarrier`
F-C14	Latest ready date for a deadline	latest date d with ETA(d) ≤ deadline (search backwards)	`latestReadyDateFor`
F-C15	Eligible	F-C2 AND F-C9 (if deadline) AND F-C11 (if budget) AND not excluded	`evaluateCarrier`

5.1 Ranking by priority

Priority	Rank eligible carriers by	Then tie-break
Cheapest	lowest F-C1	on-time ↓, cost ↑, transit ↑, name A–Z
Fastest	fewest transit days	same
Most reliable	highest on-time %	same
Balanced	highest Balanced score (below)	same

5.2 Balanced score

```
For each eligible carrier i (min/max taken over the eligible carriers):
  costScore_i  = (maxCost − cost_i)       / (maxCost − minCost)      (1 if all equal)
  speedScore_i = (maxDays − days_i)       / (maxDays − minDays)
  relScore_i   = (onTime_i − minOnTime)   / (maxOnTime − minOnTime)
  score_i      = 0.4·costScore + 0.3·speedScore + 0.3·relScore
  high-value / fragile goods: 0.3·cost + 0.3·speed + 0.4·reliability
```

Worked example — Germany, 3 pallets: costs A £540, B £660, C £1,020; days 5/3/2; on-time 92/95/98.

Carrier	costScore	speedScore	relScore	Score (0.4/0.3/0.3)	Score (high-value 0.3/0.3/0.4)
A	(1020−540)/480 = 1.000	(5−5)/3 = 0.000	(92−92)/6 = 0.000	0.400	0.300
B	(1020−660)/480 = 0.750	(5−3)/3 = 0.667	(95−92)/6 = 0.500	**0.650**	0.625
C	0.000	1.000	1.000	0.600	**0.700**

Scores are identical on every route because prices scale the same way, so Balanced picks Carrier B normally and Carrier C for high-value/fragile goods (when all three are eligible).

5.3 No eligible carrier — fallback order

- Budget can't be met → drop the budget, keep weight + deadline; best by priority.
- Deadline can't be met → fastest carrier that fits weight; show its ETA and F-C14.
- No carrier fits weight (F-B9 > 750 kg) → "No carrier fits"; show F-C3 for A, B and C.

6. Derivations used in chat / voice (always confirmed with the seller)

Derived field	Formula	Condition	Function
Item cartons	quantity ÷ units per carton	divides evenly	`deriveItemField / itemExactCartons`
Item quantity	units per carton × cartons	both known	`deriveItemField`
Units per carton	quantity ÷ cartons	divides evenly	`deriveItemField`
Booking cartons	Σ item cartons	all items have cartons	`deriveBookingTotals`
Declared value	Σ (quantity × unit price)	all items priced	`deriveBookingTotals`
Destination country	city → country map	city known	`CITY_TO_COUNTRY`
Gross weight	**never derived** — offer F-B4 and F-B13 as guidance only	—	–
Pallets	**never derived** — may show F-C3 per carrier as guidance	—	–

7. Parsing conversions

Input	Conversion	Function
“1,080”, “£5.00”, “620 kg”, “620 kilos”	strip separators, symbols and units	`parseNumber`
“one thousand and eighty”, “a dozen”, “1.2k”	word numbers / k-suffix → 1,080 / 12 / 1,200	`parseNumber`
“five pounds fifty”	5 + 50 ÷ 100 = 5.50	`parseNumber`
“500 g”	÷ 1000 → 0.5 kg	`(QUESTIONS Q-14)`
Excel date serial 46331	1899-12-30 + 46331 days = 05-11-2026	`parseDate`
“05/11/2026”, “5.11.2026”, “2026-11-05”	UK day-first → 2026-11-05; invalid dates (31-02) rejected	`parseDate`
“tomorrow”, “next Friday”, “5th November”, “in 3 days”	relative to today (UK); past day-month without year → next year	`parseSpokenDate`
HS “6912.00”, “6912 00”	strip separators → 691200	`normaliseHs`

8. Worked examples

8.1 Template booking BK-001 (all rules pass)

```
Items: MUG-01  1080 units × £5.00, 12/carton, 90 cartons, 0.5 kg, HS 691200, UK
F-B2 value      = 1080 × 5.00            = £5,400.00   declared £5,400  → R-02 pass (diff 0)
F-I3 packed     = 12 × 90                = 1,080       quantity 1,080   → R-03 pass
F-B3 cartons    = 90                                     booking 90       → R-04 pass
F-B9 per pallet = 620 ÷ 3                = 206.67 kg   Cheapest → A max 500 → R-05 pass
F-B4 net        = 1080 × 0.5             = 540 kg      gross 620 ≥ 540  → R-07 pass
Carrier A cost  = 180 × 3                = £540
ETA             = collect Thu 05-11-2026 + 5 working days = Thu 12-11-2026
Status GREEN · Recommended Carrier A Road economy £540.00
```

8.2 Booking with planted errors (today = 25-10-2026)

```
Booking: cartons 84, declared £5,000, gross 1,600 kg, 3 pallets, Warehouse,
         appointment No, ready 01-10-2026, Cheapest
Items: MUG-01 HS 90111 (numeric), 1080 units, 12/carton, 85 cartons, £5, 0.5 kg,
              origin blank
       PB-01 'Lithium-ion power bank', HS 850760, 10 units, 10/carton, 1 carton,
              £5, 0.3 kg, China
R-01 Red   : 90111 has 5 digits (<6) — possible lost leading zero (090111)
R-02 Red   : Σ = 5,400 + 50 = £5,450; |5,000 − 5,450| = £450 = 8.257% > 1% (£54.50)
R-03 Red   : MUG-01 12 × 85 = 1,020 ≠ 1,080 → cartons should be 90, or quantity 1,020
R-04 Red   : 84 ≠ 85 + 1 = 86
R-05 Red   : 1,600 ÷ 3 = 533.33 > 500 (Carrier A)
             fix: ⌈1600/500⌉ = 4 pallets (£720) or Carrier B (£660)
R-06 Red   : PB-01 matches 'lithium', 'power bank', HS 850760
R-07 pass  : net = 540 + 3 = 543 ≤ 1,600
R-08 Amber : Warehouse and no appointment
R-09 Amber : MUG-01 has no country of origin
R-10 Amber : 01-10-2026 is 24 days before 25-10-2026
Status RED · 'Recommended once fixed': Carrier B Road standard £660.00
             (A excluded: 533.33 kg > 500)
```

9. What changes if something is changed mid-way

Procedure (always): apply the change → update *derived* dependents (and say so) but only *ask* about *stated* ones → re-run `runAll()` on the whole booking → `diffResults(before, after)` → show fields changed, totals changed, issues now failing / now fixed, status, carrier, cost and ETA changes → void any earlier confirmation.

9.1 Dependency graph (input → formulas → rules)

```
quantity ──────► F-I1 line value ─► F-B2 ─► F-B5/F-B6 ─► R-02, A-07
           ├──► F-I2 net weight ─► F-B4 ─► F-B10 ─► R-07, A-03
           └──► F-I3/F-I4 ──────────────────────────► R-03      (and F-B1 → cost per unit)
unit price ─────► F-I1 ─► F-B2 ─► R-02, A-07
units/carton ──► F-I3, F-I5..F-I9 ─► R-03
item cartons ──► F-I3 ─► R-03;  F-B3 ─► F-B8 ─► R-04;  F-B11
unit weight ──► F-I2 ─► F-B4 ─► R-07, A-03
HS code ──────► F-I10 ─► R-01, A-06;  DG HS hints ─► R-06
description ──► DG keywords ─► R-06, A-09;  A-01
origin (item) ─► R-09, A-02
booking cartons ► F-B8 ─► R-04
declared value ─► F-B5/F-B6/F-B7 ─► R-02, A-07
gross weight ──► F-B9 ─► F-C2/F-C3/F-C5 ─► R-05, eligibility;  F-B10 ─► R-07, A-03;  F-C13
pallets ──────► F-B9 ─► R-05;  F-C1 cost (all carriers) ─► F-C11 budget, ranking;  F-B11
destination ──► route carriers ─► F-C1 prices ─► ranking, R-05
ready date ────► F-C6/F-C7 ETA ─► F-C9 deadline ─► eligibility;  R-10, A-05
priority ──────► ranking, priority pick ─► R-05 carrier
dest. type / appointment ─► R-08, A-04
deadline / budget / high-value / exclusions ─► eligibility, Balanced weights ─► recommendation
```

9.2 Change-impact table

Changed input	Re-computed	Rules re-checked	Recommendation
Pallets	F-B9, F-B11, F-C1 (all), F-C3, F-C5, F-C11, F-C13	R-05, A-03	Cost changes; carriers may (un)lock; may switch
Gross weight	F-B9, F-B10, F-C2, F-C3, F-C13	R-05, R-07, A-03	Eligibility may change
Booking cartons	F-B8	R-04	—
Declared value	F-B5, F-B6	R-02, A-07	—
Item quantity	F-I1, F-I2, F-I3, F-I4, F-B1, F-B2, F-B4	R-02, R-03, R-07 (+R-04 if cartons derived)	Cost per unit only
Item unit price	F-I1, F-B2, F-B7	R-02, A-07	—
Units per carton	F-I3–F-I9	R-03 (+R-04 if cartons derived)	—
Item cartons	F-I3, F-B3, F-B8, F-B11	R-03, R-04	—
Item unit weight	F-I2, F-B4, F-B10	R-07, A-03	—
HS code	F-I10, DG hints	R-01, R-06, A-06	—
Description	DG keywords (+ Gemini)	R-06, A-01, A-09	—
Country of origin	—	R-09, A-02	—
Add / remove item	all F-B sums	R-01–R-04, R-06, R-07, R-09 (V-06 if none)	Cost per unit only
Destination city/country	route; F-C1 prices	V-04, R-05	New prices; may switch
Destination type	—	R-08, A-04	—
Appointment booked / date	F-C8, F-C9	R-08, A-04	Deadline may exclude carriers
Ready date	F-C6, F-C7, F-C9, F-C14	R-10, A-05	ETAs; deadline eligibility
Priority	ranking; Balanced score	R-05 (priority pick)	Pick may change
Deadline / budget / high-value / exclusions	F-C8–F-C11, Balanced weights	—	Eligibility / pick
Seller chooses carrier	F-C1, F-C7 for that carrier	R-05 against it	Becomes booked carrier

9.3 Worked mid-way change — “add a pallet”

```
Before: Frankfurt, Cheapest, 3 pallets, 1,600 kg, items as BK-001
  F-B9 = 1600 ÷ 3 = 533.33 kg  → R-05 Red vs Carrier A (500)
  Eligible: B only (750) → Recommended once fixed: B = 220 × 3 = £660, ETA Tue 10-11-2026
Change: pallets 3 → 4
  F-B9 = 1600 ÷ 4 = 400 kg     → fits A (500), B (750), C (400, equal = pass)
  F-C1: A 180×4 = £720 · B 220×4 = £880 · C 340×4 = £1,360
  R-05 now passes; status RED → GREEN
  Recommendation B → A (£720), ETA 10-11 → Thu 12-11-2026
  Tell the seller: B on the original 3 pallets (£660) was cheaper than A on 4 (£720).
```

9.4 Worked mid-way change — “quantity 1,080 → 1,200”

```
Item cartons (provenance = derived): 1200 ÷ 12 = 100 → updated, seller told
Booking cartons (stated 90) → NOT changed → R-04 fails (90 ≠ 100)
                → ask 'Change total cartons to 100?'
Declared value (stated £5,400) → F-B2 = 1200 × 5 = £6,000 → diff £600 (10%) → R-02 Red
                → ask 'Update declared value to £6,000?'
Net weight 600 kg; gross 620 ≥ 600 → R-07 pass; packaging only 20 kg < 45 → A-03 tip
Pallets/gross unchanged → recommendation unchanged
```

10. Constants

Constant	Value	Used in
VALUE_TOLERANCE_PCT	1	R-02
HS_MIN_DIGITS	6	R-01
HS_PAD_5_DIGIT_NUMERIC	false	R-01
TRANSIT_DAY_MODE	business	F-C7
COLLECT_NEXT_WORKING_DAY_IF_WEEKEND	true	F-C6
TIMEZONE	Europe/London	R-10, F-C6
R05_MODE	priority_pick	R-05
BALANCED_WEIGHTS	cost 0.4 · speed 0.3 · reliability 0.3 (high-value 0.3/0.3/0.4)	5.2
MIN_TARE_PER_PALLET_KG	15	F-B13, A-03, R-07 fix
MIN_DESCRIPTION_CHARS	8	A-01